

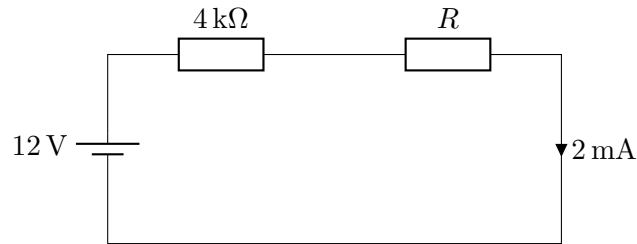
# Resistor Circuits

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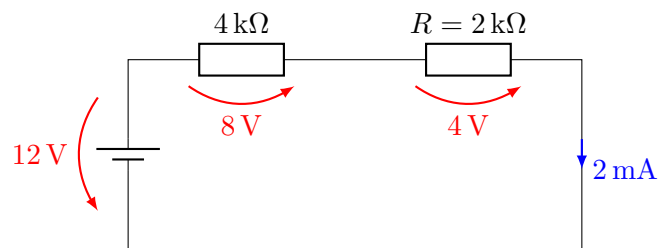
## 1 Resistor Circuit A: Series Resistors

Applying the rules for voltage and current dividers, find the resistance  $R$  in the circuit below.



### Solution

The diagram shows all currents (blue) and voltages (red) in the circuit.



Both resistors carry the same current  $I$ , so their resistances add up to the total resistance of the circuit:

$$R_a + R = \frac{V}{I} \quad (1)$$

where  $R_a = 4\text{ k}\Omega$ . Solving for  $R$  yields

$$R = \frac{V}{I} - R_a \quad (2)$$

$$= \frac{12\text{ V}}{2\text{ mA}} - 4\text{ k}\Omega \quad (3)$$

$$= \boxed{2\text{ k}\Omega} \quad (4)$$

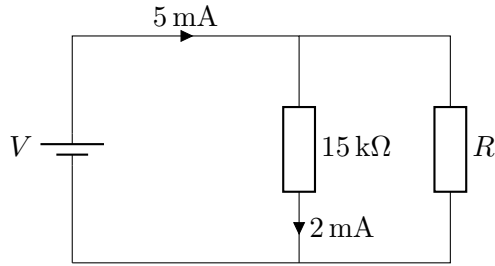
Check with the voltage divider rule: the voltage across  $R_a$  is  $IR_a = 8\text{ V}$ , leaving  $4\text{ V}$  for  $R$ . The voltages are split in the ratio  $2 : 1$ , and so are the resistances.

### Short Solution

$2\text{ k}\Omega$

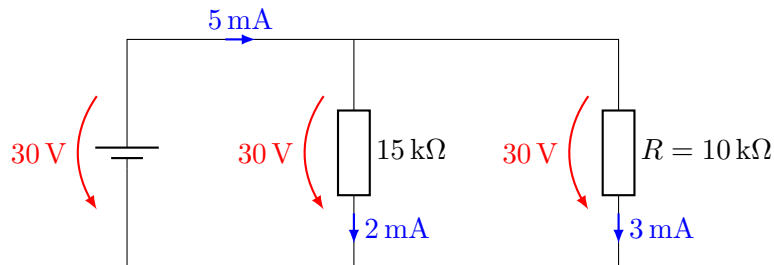
## 2 Resistor Circuit B: Parallel Resistors

Applying the rules for voltage and current dividers, find the voltage  $V$  of the battery and the resistance  $R$  in the circuit below.



### Solution

The diagram shows all currents (blue) and voltages (red) in the circuit.



Both resistors are connected directly to the battery, so the battery voltage equals the voltage across  $R_a = 15\text{ k}\Omega$ :

$$V = I_a R_a \quad (1)$$

$$= 2\text{ mA} \times 15\text{ k}\Omega \quad (2)$$

$$= \boxed{30\text{ V}} \quad (3)$$

The total current splits at the junction, so the current through  $R$  is

$$I_R = I - I_a = 3\text{ mA} \quad (4)$$

According to the current divider rule, the currents are inversely proportional to the resistances,  $I_R/I_a = R_a/R$ . Therefore

$$R = \frac{I_a}{I - I_a} R_a \quad (5)$$

$$= \frac{2\text{ mA}}{5\text{ mA} - 2\text{ mA}} \times 15\text{ k}\Omega \quad (6)$$

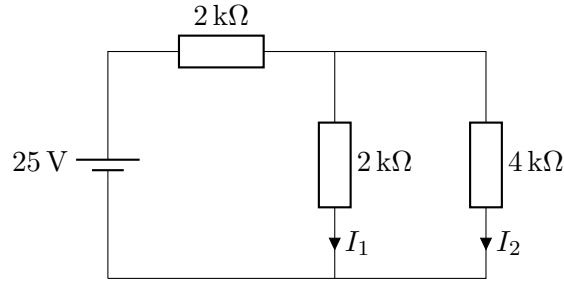
$$= \boxed{10\text{ k}\Omega} \quad (7)$$

### Short Solution

30 V, 10 kΩ

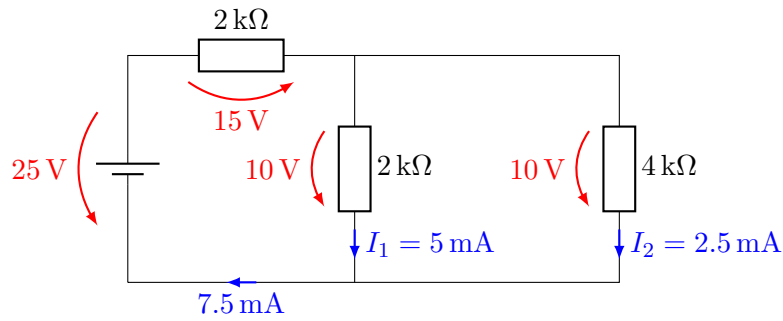
### 3 Resistor Circuit C: Voltage Divider with Parallel Load

Applying the rules for voltage and current dividers, find the currents  $I_1$  and  $I_2$  in the circuit below.



#### Solution

The diagram shows all currents (blue) and voltages (red) in the circuit.



Let  $R_a = 2\text{ k}\Omega$  be the resistor in series with the battery, and  $R_1 = 2\text{ k}\Omega$  and  $R_2 = 4\text{ k}\Omega$  the two resistors in parallel. The parallel combination has the resistance

$$R_p = \left( \frac{1}{R_1} + \frac{1}{R_2} \right)^{-1} \quad (1)$$

$$= \left( \frac{1}{2\text{ k}\Omega} + \frac{1}{4\text{ k}\Omega} \right)^{-1} \quad (2)$$

$$= \frac{4}{3}\text{ k}\Omega \quad (3)$$

$R_a$  and  $R_p$  form a voltage divider. The voltage across the parallel resistors is

$$V_p = \frac{R_p}{R_a + R_p} V \quad (4)$$

$$= \frac{\frac{4}{3}\text{ k}\Omega}{2\text{ k}\Omega + \frac{4}{3}\text{ k}\Omega} \times 25\text{ V} \quad (5)$$

$$= \frac{\frac{4}{3}}{\frac{6}{3} + \frac{4}{3}} \times 25\text{ V} = \frac{\frac{4}{3}}{\frac{10}{3}} \times 25\text{ V} = \frac{4}{10} \times 25\text{ V} \quad (6)$$

$$= 10\text{ V} \quad (7)$$

Both parallel resistors are at the voltage  $V_p$ , so the currents follow from Ohm's law:

$$I_1 = \frac{V_p}{R_1} \quad (8)$$

$$= \frac{10 \text{ V}}{2 \text{ k}\Omega} \quad (9)$$

$$= \boxed{5 \text{ mA}} \quad (10)$$

$$I_2 = \frac{V_p}{R_2} \quad (11)$$

$$= \frac{10 \text{ V}}{4 \text{ k}\Omega} \quad (12)$$

$$= \boxed{2.5 \text{ mA}} \quad (13)$$

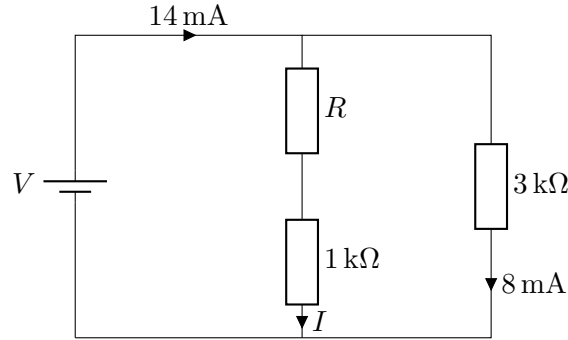
As expected from the current divider rule, the smaller resistor carries twice the current of the larger one. Together, the two currents give the current  $I = 7.5 \text{ mA}$  through  $R_a$ , which drops a voltage of  $IR_a = 15 \text{ V}$ . Together with  $V_p$ , that adds up to the battery voltage.

### Short Solution

$$I_1 = 5 \text{ mA}, I_2 = 2.5 \text{ mA}$$

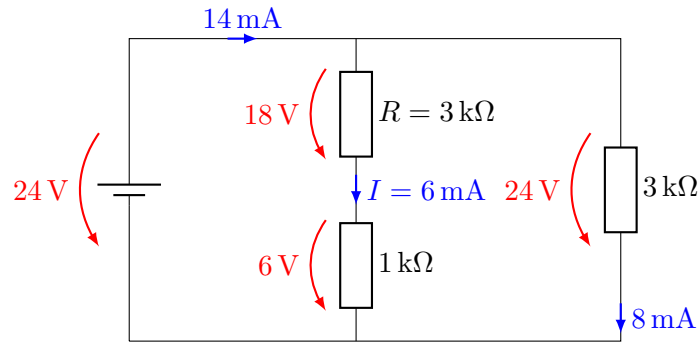
## 4 Resistor Circuit D: Parallel Branches

Applying the rules for voltage and current dividers, find the voltage  $V$  of the battery, the resistance  $R$  and the current  $I$  in the circuit below.



### Solution

The diagram shows all currents (blue) and voltages (red) in the circuit.



The right branch with  $R_b = 3\text{ k}\Omega$  is connected directly to the battery, so the battery voltage is

$$V = I_b R_b \quad (1)$$

$$= 8\text{ mA} \times 3\text{ k}\Omega \quad (2)$$

$$= \boxed{24\text{ V}} \quad (3)$$

The total current  $I_{\text{tot}}$  splits into the two branches. The current in the left branch is

$$I = I_{\text{tot}} - I_b \quad (4)$$

$$= 14\text{ mA} - 8\text{ mA} \quad (5)$$

$$= \boxed{6\text{ mA}} \quad (6)$$

The left branch consists of  $R$  and  $R_a = 1\text{ k}\Omega$  in series and is also connected to the full battery voltage,  $V = I(R + R_a)$ . Hence

$$R = \frac{V}{I} - R_a = \frac{I_b R_b}{I_{\text{tot}} - I_b} - R_a \quad (7)$$

$$= \frac{8\text{ mA} \times 3\text{ k}\Omega}{14\text{ mA} - 8\text{ mA}} - 1\text{ k}\Omega \quad (8)$$

$$= \boxed{3\text{ k}\Omega} \quad (9)$$

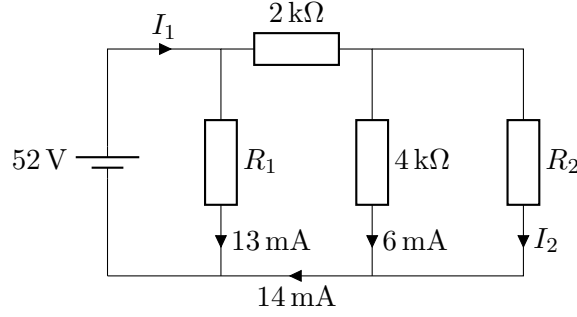
According to the voltage divider rule, the battery voltage is split in the ratio  $R : R_a = 3 : 1$  in the left branch, i.e. 18 V across  $R$  and 6 V across  $R_a$ .

### Short Solution

$$V = 24\text{ V}, R = 3\text{ k}\Omega, I = 6\text{ mA}$$

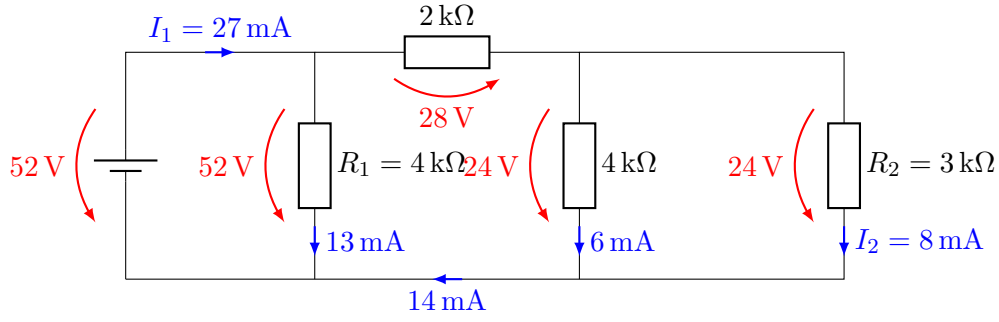
## 5 Resistor Circuit E: Three Branches

Applying the rules for voltage and current dividers, find the resistances  $R_1$  and  $R_2$  and the currents  $I_1$  and  $I_2$  in the circuit below.



### Solution

The diagram shows all currents (blue) and voltages (red) in the circuit.



$R_1$  is connected directly to the battery. With the current  $I_a = 13 \text{ mA}$  through it we find

$$R_1 = \frac{V}{I_a} \quad (1)$$

$$= \frac{52 \text{ V}}{13 \text{ mA}} \quad (2)$$

$$= \boxed{4 \text{ k}\Omega} \quad (3)$$

The current  $I_c = 14 \text{ mA}$  in the bottom wire is the current through  $R_a = 2 \text{ k}\Omega$ , i.e. the current that flows into the right part of the circuit. At the left junction the battery current splits into  $I_a$  and  $I_c$ :

$$I_1 = I_a + I_c \quad (4)$$

$$= 13 \text{ mA} + 14 \text{ mA} \quad (5)$$

$$= \boxed{27 \text{ mA}} \quad (6)$$

$I_c$  splits again into the currents through  $R_b = 4 \text{ k}\Omega$  and  $R_2$ :

$$I_2 = I_c - I_b \quad (7)$$

$$= 14 \text{ mA} - 6 \text{ mA} \quad (8)$$

$$= \boxed{8 \text{ mA}} \quad (9)$$

$R_b$  and  $R_2$  are in parallel and share the voltage  $V_b = I_b R_b$ . Hence

$$R_2 = \frac{V_b}{I_2} = \frac{I_b R_b}{I_c - I_b} \quad (10)$$

$$= \frac{6 \text{ mA} \times 4 \text{ k}\Omega}{14 \text{ mA} - 6 \text{ mA}} \quad (11)$$

$$= \boxed{3 \text{ k}\Omega} \quad (12)$$

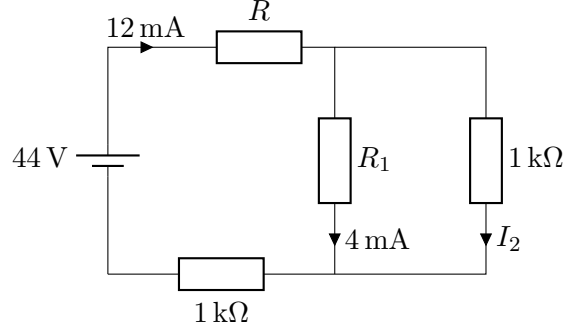
Check with the voltage divider rule: the voltage across  $R_a$  is  $V_a = I_c R_a = 28\text{ V}$ , and  $V_b = 24\text{ V}$ . Together they give the battery voltage,  $V_a + V_b = 52\text{ V}$ .

### Short Solution

$$R_1 = 4\text{ k}\Omega, R_2 = 3\text{ k}\Omega, I_1 = 27\text{ mA}, I_2 = 8\text{ mA}$$

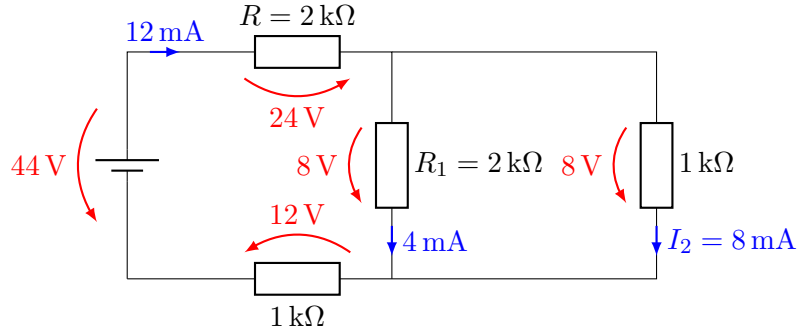
## 6 Resistor Circuit F: Parallel Pair between Two Series Resistors

Applying the rules for voltage and current dividers, find the resistances  $R$  and  $R_1$  and the current  $I_2$  in the circuit below.



### Solution

The diagram shows all currents (blue) and voltages (red) in the circuit.



The battery current  $I = 12 \text{ mA}$  splits into the currents through  $R_1$  and through the parallel resistor  $R_b = 1 \text{ k}\Omega$ :

$$I_2 = I - I_a \quad (1)$$

$$= 12 \text{ mA} - 4 \text{ mA} \quad (2)$$

$$= \boxed{8 \text{ mA}} \quad (3)$$

$R_1$  and  $R_b$  share the voltage  $V_p = I_2 R_b = (I - I_a) R_b = 8 \text{ V}$ . Hence

$$R_1 = \frac{V_p}{I_a} = \frac{I - I_a}{I_a} R_b \quad (4)$$

$$= \frac{12 \text{ mA} - 4 \text{ mA}}{4 \text{ mA}} \times 1 \text{ k}\Omega \quad (5)$$

$$= \boxed{2 \text{ k}\Omega} \quad (6)$$

The full current  $I$  flows through  $R$ , the parallel pair and  $R_c = 1 \text{ k}\Omega$  in series, so the battery voltage is divided as

$$V = IR + V_p + IR_c \quad (7)$$

Solving for  $R$ :

$$R = \frac{V - IR_c - (I - I_a)R_b}{I} \quad (8)$$

$$= \frac{44 \text{ V} - 12 \text{ mA} \times 1 \text{ k}\Omega - (12 \text{ mA} - 4 \text{ mA}) \times 1 \text{ k}\Omega}{12 \text{ mA}} \quad (9)$$

$$= \boxed{2 \text{ k}\Omega} \quad (10)$$

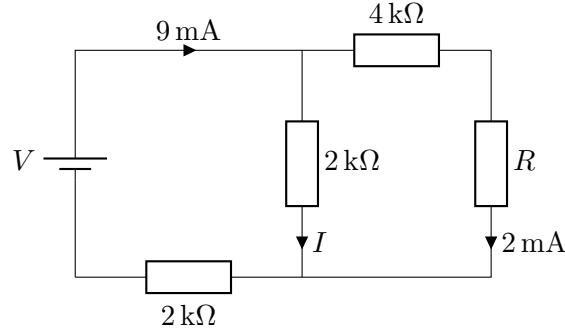
The voltages across  $R$ , the parallel pair and  $R_c$  are 24 V, 8 V and 12 V, which add up to 44 V.

**Short Solution**

$$R = 2\text{ k}\Omega, R_1 = 2\text{ k}\Omega, I_2 = 8\text{ mA}$$

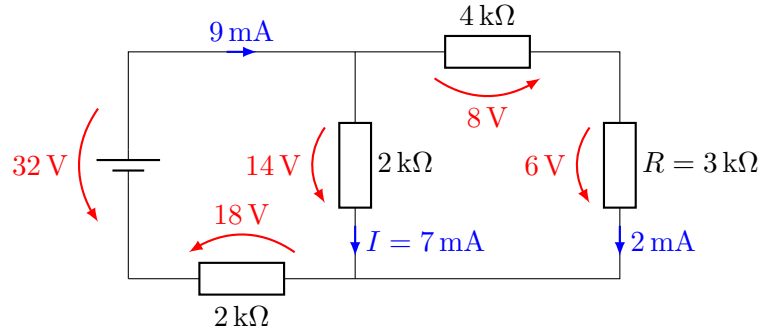
## 7 Resistor Circuit G: Branch with Two Series Resistors

Applying the rules for voltage and current dividers, find the voltage  $V$  of the battery, the resistance  $R$  and the current  $I$  in the circuit below.



### Solution

The diagram shows all currents (blue) and voltages (red) in the circuit.



The total current  $I_{\text{tot}} = 9 \text{ mA}$  splits into the current  $I$  through  $R_b = 2 \text{ k}\Omega$  and the current  $I_c = 2 \text{ mA}$  through the right branch:

$$I = I_{\text{tot}} - I_c \quad (1)$$

$$= 9 \text{ mA} - 2 \text{ mA} \quad (2)$$

$$= \boxed{7 \text{ mA}} \quad (3)$$

Both branches share the voltage  $V_p = IR_b = (I_{\text{tot}} - I_c)R_b = 14 \text{ V}$ . The right branch consists of  $R_c = 4 \text{ k}\Omega$  and  $R$  in series, so  $V_p = I_c(R_c + R)$  and

$$R = \frac{V_p}{I_c} - R_c = \frac{I_{\text{tot}} - I_c}{I_c} R_b - R_c \quad (4)$$

$$= \frac{9 \text{ mA} - 2 \text{ mA}}{2 \text{ mA}} \times 2 \text{ k}\Omega - 4 \text{ k}\Omega \quad (5)$$

$$= \boxed{3 \text{ k}\Omega} \quad (6)$$

The battery voltage is the sum of the voltage across  $R_a = 2 \text{ k}\Omega$ , which carries the total current, and the voltage across the parallel part:

$$V = I_{\text{tot}}R_a + (I_{\text{tot}} - I_c)R_b \quad (7)$$

$$= 9 \text{ mA} \times 2 \text{ k}\Omega + (9 \text{ mA} - 2 \text{ mA}) \times 2 \text{ k}\Omega \quad (8)$$

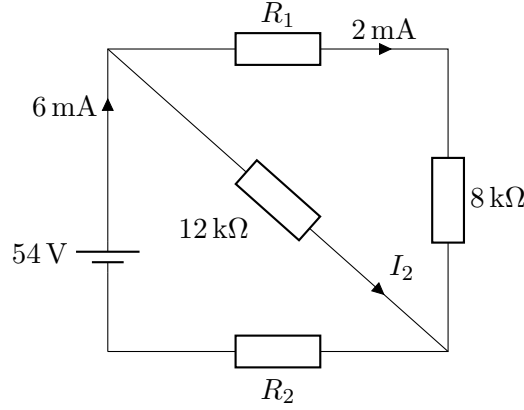
$$= \boxed{32 \text{ V}} \quad (9)$$

### Short Solution

$$V = 32 \text{ V}, R = 3 \text{ k}\Omega, I = 7 \text{ mA}$$

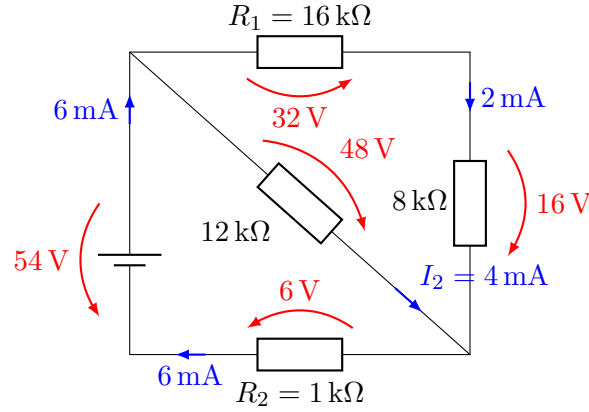
## 8 Resistor Circuit H: Diagonal Resistor

Applying the rules for voltage and current dividers, find the resistances  $R_1$  and  $R_2$  and the current  $I_2$  in the circuit below.



### Solution

The diagram shows all currents (blue) and voltages (red) in the circuit.



The diagonal resistor  $R_b = 12 \text{ k}\Omega$  is connected in parallel with the series combination of  $R_1$  and  $R_a = 8 \text{ k}\Omega$ . This parallel part is in series with  $R_2$ , which carries the full battery current  $I = 6 \text{ mA}$ .

The battery current splits into the current  $I_a = 2 \text{ mA}$  through the upper path and the current through the diagonal:

$$I_2 = I - I_a \quad (1)$$

$$= 6 \text{ mA} - 2 \text{ mA} \quad (2)$$

$$= \boxed{4 \text{ mA}} \quad (3)$$

The voltage across the parallel part is  $V_p = I_2 R_b = (I - I_a) R_b = 48 \text{ V}$ . The upper path carries  $I_a$  at the same voltage,  $V_p = I_a (R_1 + R_a)$ , so

$$R_1 = \frac{V_p}{I_a} - R_a = \frac{I - I_a}{I_a} R_b - R_a \quad (4)$$

$$= \frac{6 \text{ mA} - 2 \text{ mA}}{2 \text{ mA}} \times 12 \text{ k}\Omega - 8 \text{ k}\Omega \quad (5)$$

$$= \boxed{16 \text{ k}\Omega} \quad (6)$$

The remaining battery voltage  $V - V_p$  drops across  $R_2$ :

$$R_2 = \frac{V - V_p}{I} = \frac{V - (I - I_a)R_b}{I} \quad (7)$$

$$= \frac{54 \text{ V} - (6 \text{ mA} - 2 \text{ mA}) \times 12 \text{ k}\Omega}{6 \text{ mA}} \quad (8)$$

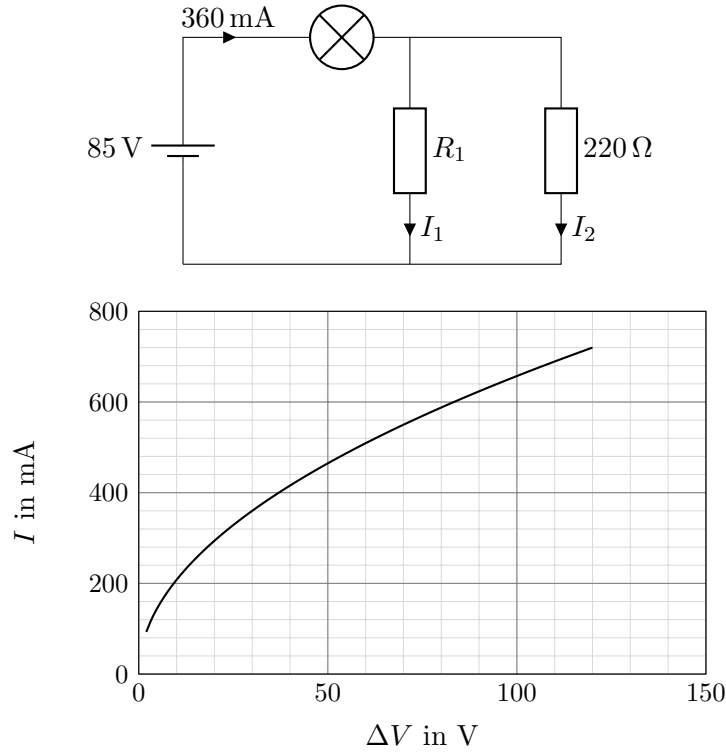
$$= \boxed{1 \text{ k}\Omega} \quad (9)$$

### Short Solution

$$R_1 = 16 \text{ k}\Omega, R_2 = 1 \text{ k}\Omega, I_2 = 4 \text{ mA}$$

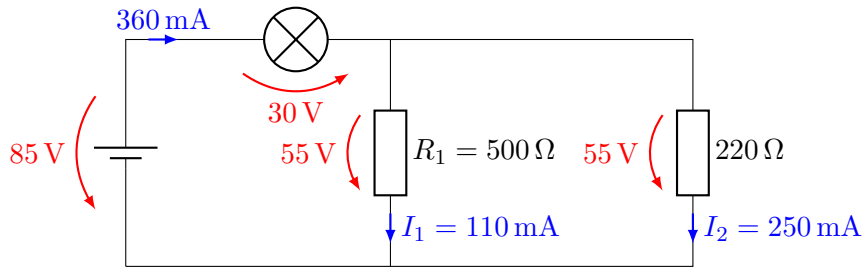
## 9 Resistor Circuit I: Light Bulb

A light bulb is connected in series with two resistors in parallel. Using the characteristic of the light bulb shown below, find the resistance  $R_1$  and the currents  $I_1$  and  $I_2$ .



### Solution

The diagram shows all currents (blue) and voltages (red) in the circuit.



The full current  $I = 360\text{ mA}$  flows through the light bulb. From the characteristic we read the voltage across the bulb at this current:

$$V_L \approx 30\text{ V} \quad (1)$$

According to the voltage divider rule, the rest of the battery voltage drops across the two parallel resistors. Their common voltage is

$$V_p = V - V_L = 55\text{ V} \quad (2)$$

The current through  $R_b = 220\ \Omega$  follows from Ohm's law:

$$I_2 = \frac{V_p}{R_b} = \frac{V - V_L}{R_b} \quad (3)$$

$$= \frac{85\text{ V} - 30\text{ V}}{220\ \Omega} \quad (4)$$

$$= \boxed{250\text{ mA}} \quad (5)$$

The rest of the current flows through  $R_1$ :

$$I_1 = I - I_2 = I - \frac{V - V_L}{R_b} \quad (6)$$

$$= 360 \text{ mA} - \frac{85 \text{ V} - 30 \text{ V}}{220 \Omega} \quad (7)$$

$$= \boxed{110 \text{ mA}} \quad (8)$$

Finally,

$$R_1 = \frac{V_p}{I_1} = \frac{V - V_L}{I - \frac{V - V_L}{R_b}} \quad (9)$$

$$= \frac{85 \text{ V} - 30 \text{ V}}{360 \text{ mA} - \frac{85 \text{ V} - 30 \text{ V}}{220 \Omega}} \quad (10)$$

$$= \boxed{500 \Omega} \quad (11)$$

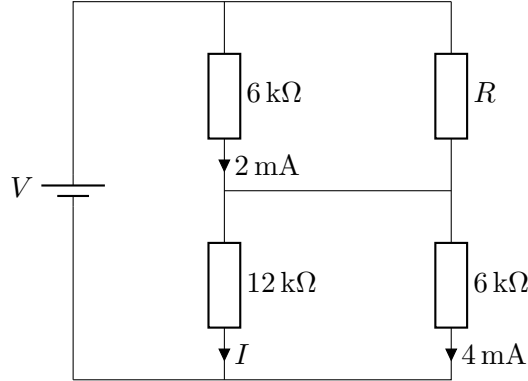
Because the voltage is read from a graph, results that differ slightly from these values are acceptable.

### Short Solution

$$R_1 = 500 \Omega, I_1 = 110 \text{ mA}, I_2 = 250 \text{ mA}$$

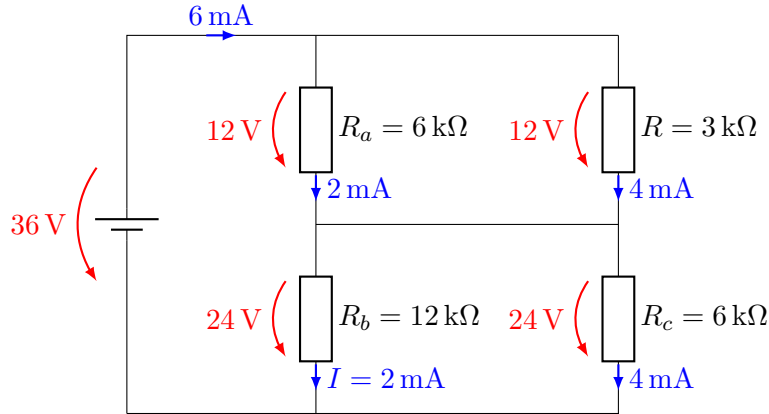
## 10 Resistor Circuit J: Two Parallel Pairs in Series

Applying the rules for voltage and current dividers, find the voltage  $V$  of the battery, the resistance  $R$  and the current  $I$  in the circuit below.



### Solution

The diagram shows all currents (blue) and voltages (red) in the circuit.



The circuit consists of two parallel pairs connected in series. In the upper pair,  $R_a = 6\text{ k}\Omega$  carries the current  $I_a = 2\text{ mA}$ . In the lower pair,  $R_c = 6\text{ k}\Omega$  carries the current  $I_c = 4\text{ mA}$ .

Both resistors of a pair are at the same voltage. The voltages across the upper and the lower pair follow from Ohm's law:

$$V_1 = I_a R_a = 2\text{ mA} \times 6\text{ k}\Omega = 12\text{ V} \quad (1)$$

$$V_2 = I_c R_c = 4\text{ mA} \times 6\text{ k}\Omega = 24\text{ V} \quad (2)$$

The two pairs are in series, so their voltages add up to the battery voltage:

$$V = V_1 + V_2 \quad (3)$$

$$= 12\text{ V} + 24\text{ V} \quad (4)$$

$$= \boxed{36\text{ V}} \quad (5)$$

The current through  $R_b = 12\text{ k}\Omega$  follows from the voltage across the lower pair:

$$I = \frac{V_2}{R_b} \quad (6)$$

$$= \frac{24\text{ V}}{12\text{ k}\Omega} \quad (7)$$

$$= \boxed{2\text{ mA}} \quad (8)$$

The battery current flows through both pairs. At the lower pair it splits into  $I$  and  $I_c$ :

$$I_{\text{tot}} = I + I_c = 2 \text{ mA} + 4 \text{ mA} = 6 \text{ mA} \quad (9)$$

At the upper pair, the part that does not flow through  $R_a$  flows through  $R$ :

$$I_R = I_{\text{tot}} - I_a = 6 \text{ mA} - 2 \text{ mA} = 4 \text{ mA} \quad (10)$$

$R$  is at the voltage  $V_1$  of the upper pair. Hence

$$R = \frac{V_1}{I_R} \quad (11)$$

$$= \frac{12 \text{ V}}{4 \text{ mA}} \quad (12)$$

$$= \boxed{3 \text{ k}\Omega} \quad (13)$$

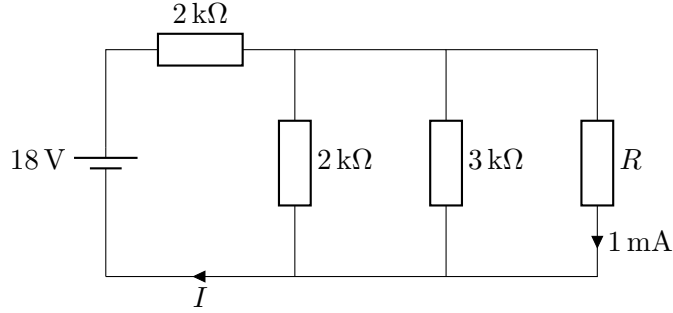
Check with the current divider rule: in the upper pair,  $R$  carries twice the current of  $R_a$  and has half its resistance.

### Short Solution

$$V = 36 \text{ V}, R = 3 \text{ k}\Omega, I = 2 \text{ mA}$$

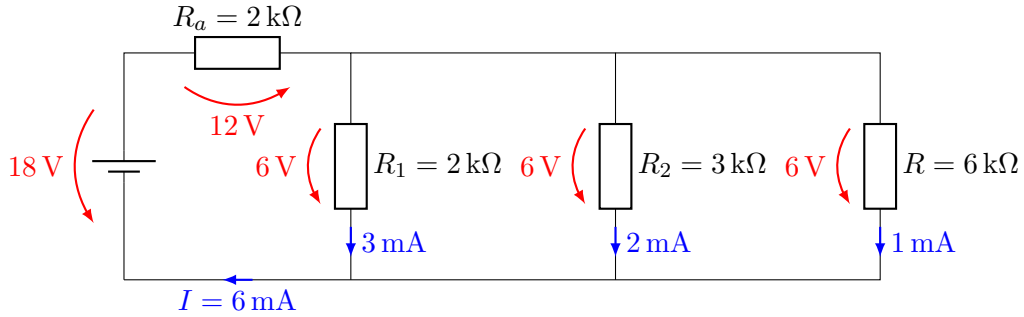
## 11 Resistor Circuit K: Three Resistors in Parallel

Applying the rules for voltage and current dividers, find the voltage  $V_p$  across the parallel resistors, the resistance  $R$  and the battery current  $I$  in the circuit below.



### Solution

The diagram shows all currents (blue) and voltages (red) in the circuit.



Let  $R_a = 2\text{ k}\Omega$  be the resistor in series with the battery, and  $R_1 = 2\text{ k}\Omega$  and  $R_2 = 3\text{ k}\Omega$  the two known resistors in parallel. The unknown resistor  $R$  carries the current  $I_R = 1\text{ mA}$ .

Neither the battery current nor the voltage  $V_p$  is known. We therefore use the junction rule: the current through  $R_a$  equals the sum of the currents in the three parallel branches. With Ohm's law, and since  $R_a$  is at the voltage  $V - V_p$ ,

$$\frac{V - V_p}{R_a} = \frac{V_p}{R_1} + \frac{V_p}{R_2} + I_R \quad (1)$$

Collecting the terms with  $V_p$  on one side yields

$$V_p = \frac{\frac{V}{R_a} - I_R}{\frac{1}{R_a} + \frac{1}{R_1} + \frac{1}{R_2}} \quad (2)$$

$$= \frac{\frac{18\text{ V}}{2\text{ k}\Omega} - 1\text{ mA}}{\frac{1}{2\text{ k}\Omega} + \frac{1}{2\text{ k}\Omega} + \frac{1}{3\text{ k}\Omega}} \quad (3)$$

$$= \frac{9\text{ mA} - 1\text{ mA}}{\frac{3}{6\text{ k}\Omega} + \frac{3}{6\text{ k}\Omega} + \frac{2}{6\text{ k}\Omega}} = \frac{8\text{ mA}}{\frac{8}{6\text{ k}\Omega}} \quad (4)$$

$$= \boxed{6\text{ V}} \quad (5)$$

The unknown resistor is at the same voltage, so

$$R = \frac{V_p}{I_R} \quad (6)$$

$$= \frac{6 \text{ V}}{1 \text{ mA}} \quad (7)$$

$$= \boxed{6 \text{ k}\Omega} \quad (8)$$

The remaining 12 V of the battery voltage drop across  $R_a$ :

$$I = \frac{V - V_p}{R_a} \quad (9)$$

$$= \frac{18 \text{ V} - 6 \text{ V}}{2 \text{ k}\Omega} \quad (10)$$

$$= \boxed{6 \text{ mA}} \quad (11)$$

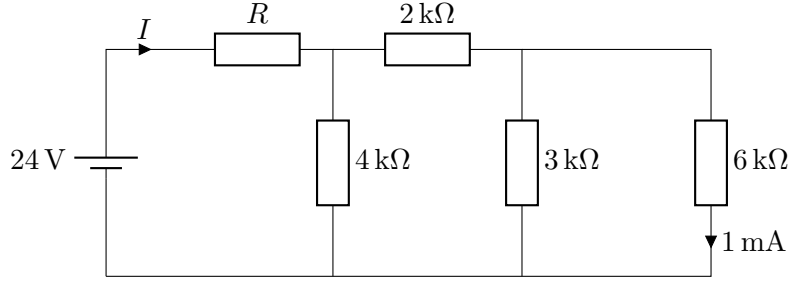
Check: the currents through  $R_1$  and  $R_2$  are  $V_p/R_1 = 3 \text{ mA}$  and  $V_p/R_2 = 2 \text{ mA}$ . Together with  $I_R$  they add up to the battery current  $I$ .

### Short Solution

$$V_p = 6 \text{ V}, R = 6 \text{ k}\Omega, I = 6 \text{ mA}$$

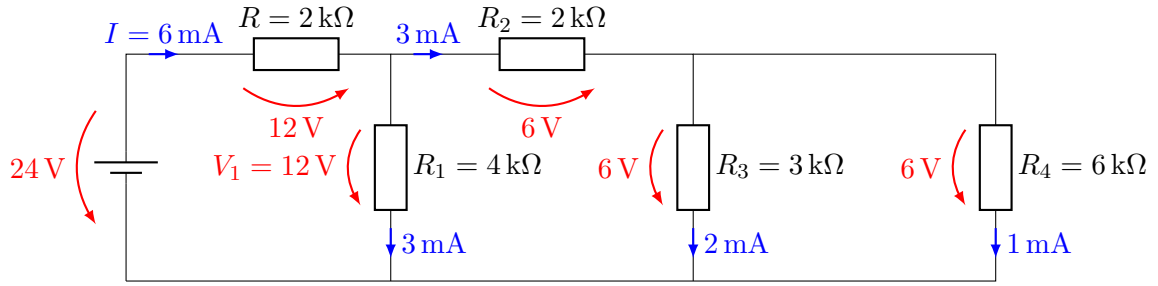
## 12 Resistor Circuit L: Unknown Series Resistor

Applying the rules for voltage and current dividers, find the resistance  $R$ , the battery current  $I$  and the voltage  $V_1$  across the  $4\text{ k}\Omega$  resistor in the circuit below.



### Solution

The diagram shows all currents (blue) and voltages (red) in the circuit.



Let  $R_1 = 4\text{ k}\Omega$  be the resistor in the middle branch. The right branch consists of  $R_2 = 2\text{ k}\Omega$  in series with the parallel pair  $R_3 = 3\text{ k}\Omega$  and  $R_4 = 6\text{ k}\Omega$ . The only given current is  $I_4 = 1\text{ mA}$  through  $R_4$ , so we start at the far right and work towards the battery.

The voltage across the parallel pair is

$$V_{34} = I_4 R_4 = 1\text{ mA} \times 6\text{ k}\Omega = 6\text{ V} \quad (1)$$

$R_3$  is at the same voltage and carries

$$I_3 = \frac{V_{34}}{R_3} = \frac{6\text{ V}}{3\text{ k}\Omega} = 2\text{ mA} \quad (2)$$

Both currents flow through  $R_2$ :

$$I_b = I_3 + I_4 = 2\text{ mA} + 1\text{ mA} = 3\text{ mA} \quad (3)$$

so the voltage across  $R_2$  is  $V_2 = I_b R_2 = 6\text{ V}$ .

The middle branch is in parallel with the whole right branch. Its voltage is

$$V_1 = V_2 + V_{34} \quad (4)$$

$$= 6\text{ V} + 6\text{ V} \quad (5)$$

$$= \boxed{12\text{ V}} \quad (6)$$

The current through  $R_1$  is  $I_1 = \frac{V_1}{R_1} = 3\text{ mA}$ , and the battery current splits into  $I_1$  and  $I_b$ :

$$I = I_1 + I_b \quad (7)$$

$$= 3\text{ mA} + 3\text{ mA} \quad (8)$$

$$= \boxed{6\text{ mA}} \quad (9)$$

The rest of the battery voltage drops across  $R$ , which carries the full battery current:

$$R = \frac{V - V_1}{I} \quad (10)$$

$$= \frac{24 \text{ V} - 12 \text{ V}}{6 \text{ mA}} \quad (11)$$

$$= \boxed{2 \text{ k}\Omega} \quad (12)$$

### Short Solution

$$R = 2 \text{ k}\Omega, I = 6 \text{ mA}, V_1 = 12 \text{ V}$$